

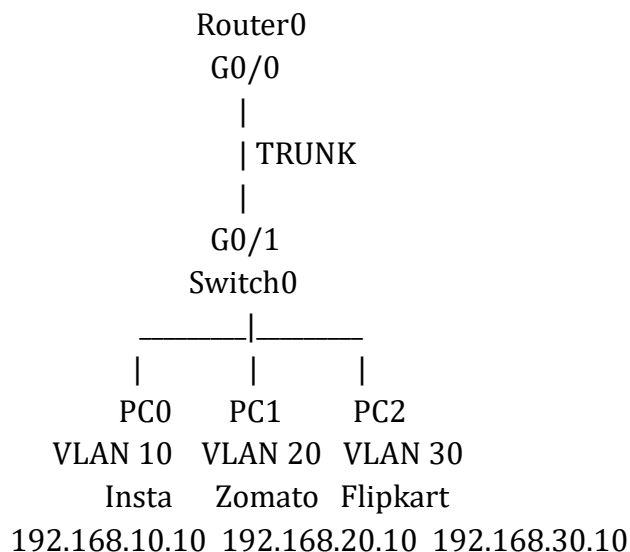
## DAY-5

Inter-VLAN Routing Concepts???? Theory: Why is Inter-VLAN Routing Needed? Problem: VLANs cannot communicate with each other Solution: Use a router or Layer 3 switch Methods of InterVLAN Routing: Router-on-a-Stick (RoS) (Single router interface for multiple VLANs) Layer 3 Switch Routing???? Practical: Router-on-a-Stick Configuration???? Hands-on Lab Enable router sub-interfaces for VLANs

**[Q-1] On a Cisco Packet Tracer or GNS3 simulation, create three VLANs named Insta, Zomato, and Flipkart on a single switch, and assign at least one PC to each VLAN.**

**(A)**

### ➤ Topology



### ➤ VLAN Details

| VLAN ID | VLAN Name | Network | Default Gateway |
|---------|-----------|---------|-----------------|
|---------|-----------|---------|-----------------|

|    |          |                 |              |
|----|----------|-----------------|--------------|
| 10 | Insta    | 192.168.10.0/24 | 192.168.10.1 |
| 20 | Zomato   | 192.168.20.0/24 | 192.168.20.1 |
| 30 | Flipkart | 192.168.30.0/24 | 192.168.30.1 |

➤ **Switch Configuration - Create VLANs**

```
enable
configure terminal
```

```
vlan 10 name
Insta exit
```

```
vlan 20 name
Zomato
exit
```

```
vlan 30 name
Flipkart exit
```

**[Q-2] Configure a Router-on-a-Stick setup by creating sub-interfaces (e.g., GigabitEthernet0/0.10, .20, .30) on a simulated router, each with an IP address for the corresponding VLANs Insta, Zomato, and Flipkart.**

**(A)**

➤ **Configure Router-on-a-Stick**

```
enable
configure terminal
```

```
interface gigabitEthernet0/0 no
shutdown
exit
```

```
! =====
```

! VLAN 10 - Insta

! =====

```
interface gigabitEthernet0/0.10
encapsulation dot1Q 10 ip address
192.168.10.1 255.255.255.0 exit
```

! =====

! VLAN 20 - Zomato

! =====

```
interface gigabitEthernet0/0.20
encapsulation dot1Q 20 ip address
192.168.20.1 255.255.255.0 exit
```

! =====

! VLAN 30 - Flipkart

! =====

```
interface gigabitEthernet0/0.30
encapsulation dot1Q 30 ip address
192.168.30.1 255.255.255.0 exit
```

```
end write
memory
```

**[Q-3] Assign the switch port connected to the router as a trunk port, and verify trunking is active using the show interfaces trunk command. Hint: Use the switchport mode trunk command on the relevant interface.**

**(A)**

➤ **Configure the Switch Trunk Port**

```
enable
configure terminal
```

```
interface gigabitEthernet0/1
switchport mode trunk no
shutdown
exit
```

end

➤ **Verify trunking:**

- show interfaces trunk

➤ **Expected output:**

- Port      Mode      Encapsulation      Status      Native vlan  
    Gi0/1    on        802.1q      trunking    1
- Port      Vlans allowed on trunk Gi0/1    1-4094
- Port      Vlans allowed and active in management domain Gi0/1    1,10,20,30

**[Q-4] Test inter-VLAN connectivity by pinging from a PC in the Insta VLAN to a PC in the Zomato VLAN, and explain why the ping succeeds or fails based on your configuration.**

**(A)**

➤ **Test Inter-VLAN Connectivity**

From **PC0 in VLAN 10**, open:

**Desktop → Command Prompt**

Ping PC1 in VLAN 20: ping  
192.168.20.10

You can also test PC2: ping  
192.168.30.10

**Expected Result**

Reply from 192.168.20.10

➤ **Why Does the Ping Succeed?**

```
PC0
VLAN 10
192.168.10.10
|
v
Switch0
|
| VLAN 10 tagged traffic
| TRUNK
v
Router G0/0.10
192.168.10.1
|
| Routes traffic between networks
v
Router G0/0.20
192.168.20.1
|
| VLAN 20 tagged traffic
v
Switch0
|
v
PC1
VLAN 20
192.168.20.10
```

**[Q-5] Refactor your Router-on-a-Stick configuration to add a new VLAN called 'Spotify' without disrupting existing VLAN communication, and document the exact commands you used for both the switch and the router.**

(A)

➤ **Add New VLAN – Spotify**

| VLAN ID | VLAN Name | Network         | Gateway      |
|---------|-----------|-----------------|--------------|
| 40      | Spotify   | 192.168.40.0/24 | 192.168.40.1 |

➤ **Step 1: Switch Configuration for Spotify**

Create VLAN 40: enable  
configure terminal

vlan 40 name  
Spotify exit

Assume the new PC is connected to FastEthernet0/4.

Configure the port:

```
interface fastEthernet0/4
switchport mode access
switchport access vlan 40 no
shutdown
exit
```

If the trunk already allows all VLANs, VLAN 40 can pass automatically.

To explicitly verify the trunk:

```
show interfaces trunk
```

If required, explicitly allow VLAN 40:

```
interface gigabitEthernet0/1
switchport trunk allowed vlan add 40
exit Save: end write memory
```

➤ **Step 2: Configure Spotify PC**

IP Address: 192.168.40.10  
Subnet Mask: 255.255.255.0

Default Gateway: 192.168.40.1

➤ **Step 3: Add Router Sub-Interface**

On Router0:

enable configure  
terminal

```
interface gigabitEthernet0/0.40
encapsulation dot1Q 40 ip address
192.168.40.1 255.255.255.0 exit
```

end write  
memory

The existing configuration remains unchanged:

G0/0.10 → Insta

G0/0.20 → Zomato

G0/0.30 → Flipkart

G0/0.40 → Spotify